

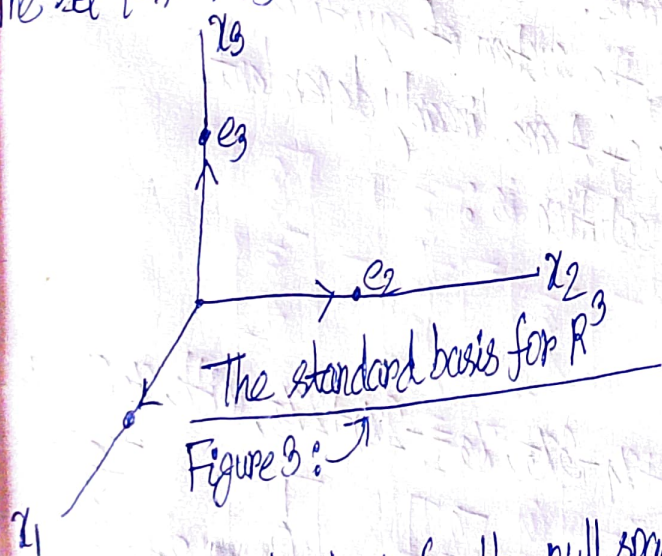
Definition: A basis for a subspace H of $\mathbb{R}^n$ is a linearly independent set in H that spans H .

Example 5: The columns of an invertible $n \times n$ matrix form a basis for all of $\mathbb{R}^n$ because they are linearly independent and span $\mathbb{R}^n$, by the Invertible Matrix Theorem.

One such matrix is the $n \times n$ identity matrix. Its columns are denoted by $e_1, \dots, e_n$.

$e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \dots, e_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$

The set $\{e_1, \dots, e_n\}$ is called the **standard basis** for $\mathbb{R}^n$. $\square$



Example 6: Find a basis for the null space of the matrix.

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

Solution: By Defn, a nullspace is the set $\text{Nul } A$ of all solutions of the equation $Ax = 0$. By Theorem 3 from Chapter 1, the matrix equation $Ax = 0$ has the same solution set as the system of equations corresponding to the augmented matrix $[A \ 0]$.

$$\begin{bmatrix} -3 & 6 & -1 & 1 & -7 & 0 \\ 1 & -2 & 2 & 3 & -1 & 0 \\ 2 & -4 & 5 & 8 & -4 & 0 \end{bmatrix} \xrightarrow[\text{Row 1 and Row 2}]{\text{Interchange}} \begin{bmatrix} 1 & -2 & 2 & 3 & -1 & 0 \\ -3 & 6 & -1 & 1 & -7 & 0 \\ 2 & -4 & 5 & 8 & -4 & 0 \end{bmatrix}$$